

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and caused reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

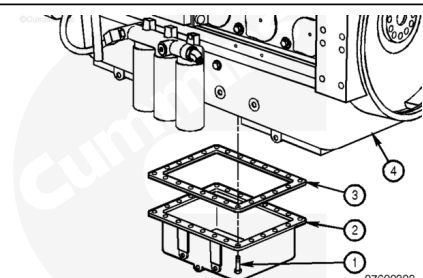
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Drain the engine oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).
- Remove the lubricating oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



The QSK45 and QSK60 engines use a variety of lubricating oil pan and cover plate configurations. **Only** single and double oil pans are used. When a double oil pan is used, **only** cover

plates will be used to cover the remainder of the oil pan adapter. Full length deep and shallow sumps are available as options.

Single oil pans and cover plates are mounted with 24 capscrews and double oil pans are mounted with 38 capscrews.

Remove the capscrews (1), oil pan (2), and gasket (3) from the oil pan adapter (4).

Inspect gaskets for reuse (3).

Clean and Inspect for Reuse

⚠ WARNING ⚠

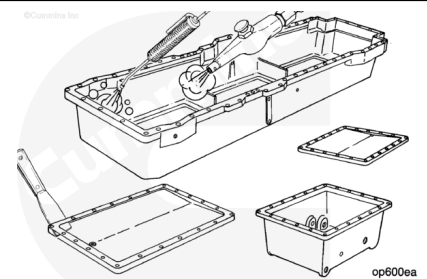
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

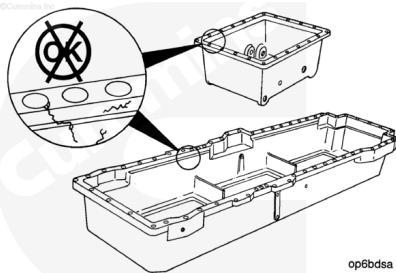


Clean all gasket surfaces on the oil pan.

Clean all parts. Use steam. Dry with compressed air.

Clean drain plug magnets. Use solvent. Dry with compressed air.

Inspect all areas of the oil pan for cracks or other damage. If any of the areas are cracked or damaged, the oil pan **must** be replaced.



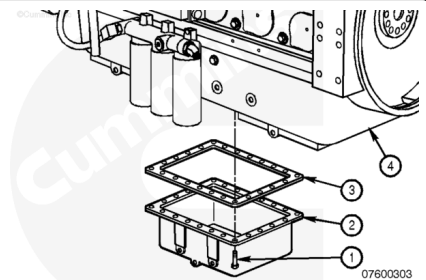
Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Do not use gasket cement on the oil pan gaskets. Gasket cement will prevent the gasket from sealing properly.



Single oil pans and cover plates are mounted with 24 capscrews and double oil pans are mounted with 38 capscrews.

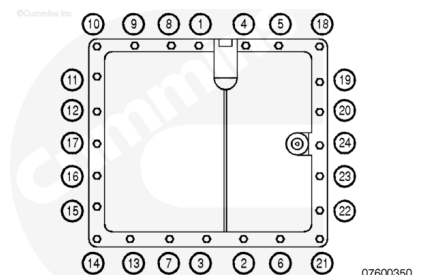
Position the oil pan (2) and gaskets (3) on the oil adapter (4) with capscrews.

Tighten the capscrews in the pattern as shown.

Torque Value: 36 n•m [26.5 ft-lb]

Install and tighten the drain plug(s).

Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

- Install new lubricating oil filters. Refer to Procedure 007-013 (/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html).
- Fill the engine with clean engine oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html).
- Operate the engine at idle speed. Inspect for engine lubricating oil leaks.

